PA194ZR

A locally certified zero-return implementation of a PA consistency machine

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13 November 2026, Pacific time

Pinned source revision: `17f92105e2bc4317feae6c219cb2143437b482ff`

Abstract

We compare the forty-one principal research packets in the supplied PA machine corpus and construct an explicit, distinct, 194-state binary single-tape machine. A verified machine with fewer than 194 states was not obtained. The result therefore addresses the requested same-size alternative, with its novelty stated precisely.

The new implementation, PA194ZR, replaces nine unconditional source returns by zero-tested decrements. The transformation changes two native transition rows of PA194CS, affects eight shared-comparator returns and one axiom-emitter exit, and preserves the entire source execution from its designated entry. The preservation proof uses a 129-site region: its only three incoming edges are zero branches of a decrement of the selected register, and the region never increments that register. This is an elementary unbounded induction, with finite graph premises verified by two independent checkers.

The complete 388-row table, deterministic reconstruction, all 4,754 universal source outcomes, a separate literal-tape symbolic check, and an exact 59,816-step blank initialization are included. Graph comparison excludes all supplied 194-state designs even under state renaming, choice of initial control, reflection and global symbol interchange. The ordinary mathematical PA proof transfers the supplied separate-head calculus calibration through exact source equivalence and positive physical simulation. A new end-to-end Lean or Rocq theorem is not claimed.

194 working states 2 symbols 1 tape 388 transitions

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(M_{194\mathrm{ZR}}, \mathrm{blank}) \iff \mathrm{Con}(\mathrm{PA}).$$

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1 Result and exact scope

The definitive file is `machine/pa194zr.tm`. Working controls are $0, \ldots, 193$, the initial control is zero, and the separate terminal halt label is 194. The tape is two-way infinite, its alphabet is $\{0, 1\}$, every initial cell is zero, and the head starts at coordinate zero. Every transition writes one bit, moves one cell left or right, and selects its next control. The terminal label has no outgoing rows and is excluded from the working-state count. There are no stationary moves, extra symbols, extra tapes, or prewritten input words.

A line `q b w D n` means that control q reading b writes w, moves in direction $D \in \{L, R\}$, and enters n. Appendix A prints the whole table. Its exact SHA-256, including the final newline, is

`11e4b81320382c3293cb075119f6f87b4b454f96e13798c432744a391ee2398f`.

The native-label version starts at 74 and halts at 254; its SHA-256 is

`745dd4343d88f6d3458cbde6146cfe360283bc7e8a9113d020eed999b81ad85d`.

The file `machine/labels.json` supplies the checked conjugacy between these two presentations. All 194 working controls are reachable in the transition graph. This is a finite graph fact, not an assertion that all controls are visited on the actual blank run.

Theorem 1.1 (Consistency equivalence)**.** *For the literal machine $M_{194\mathrm{ZR}}$ above,*

$$\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_{194\mathrm{ZR}}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

Here Con(PA) *is the absence of a finite proof of* $0 = 1$ *in a fixed ordinary recursively checked PA proof system. The proof uses the supplied ordinary mathematical calibration of the separate-head proof-search program, and the new source and physical preservation results below.*

The theorem is a claim about PA-provability; it does not assume Con(PA) as an axiom. Actual nonhalting follows if PA is consistent. The proof dependency is explicit: this work does not replace the inherited syntactic calibration with a claim that the repository already has a completed formal endpoint.

1.1 What the new design changes

PA194ZR inherits the 190-token axiom emitter and symmetric cached-head Shift implementation from PA194CS. It changes the return protocol at nine actual source sites. These are executable instructions on ordinary source paths, not edits to unused rows. The native table changes are particularly small:

Control	Read	PA194CS	PA194ZR
103	1	$(0, L, 302)$	$(1, R, 17)$
2066	1	$(0, L, 302)$	$(1, R, 17)$

Native 17 dispatches the relevant decrement of logical register R_3. One shared reader row accounts for eight source occurrences; the other accounts for the AX exit. The proof that these decrements act as returns is source-level and does not assume the values are zero on arbitrary tapes. The literal-tape contracts separately establish the behavior of the changed physical table.

This is a distinct implementation, not a new numerical axiom theory or a new proof-search architecture. The idea of replacing a jump with a known-zero decrement is inherited from earlier work. The new

result is its simultaneous use at these nine sites, the small common invariant, and a validated 194-state table whose complete initialization and physical source binding hold. The distinction is operational and verified; it is not presented as a conceptual breakthrough or a state count record below the supplied frontier.

2 Comparison of the complete supplied corpus

The comparison is pinned to `VladimirReshetnikov/ProveIt` commit

`17f92105e2bc4317feae6c219cb2143437b482ff`.

These are the requested repository locations [1, 2]. The repository has reorganized the historical submissions: 36 research packets are filed under `Computability/BusyBeaver/Research/PA/docs`, 21 of those remain as incoming archives, and five further archives have arrived. Consequently the requested comparison covers **41 principal research packets**, not just the 26 ZIPs currently visible in `docs/incoming`. This distinction matters: scanning only the incoming filenames would omit fifteen earlier packets; counting every filed copy as an independent design would overcount. The full packet ledger is included in `comparison/archive_ledger.csv` and Appendix B.

2.1 A reproducible, explicitly scoped census

The census recursively opens each distinct ZIP payload, retains every reference to it, and recognizes complete numeric, five-column binary transition tables in the supported text formats. Each recognized table has two rows for every working control and exactly one undefined terminal control. Direct PA tables and the six nested archives retained in the filed historical packets are included. The recognizer does not discover machines embedded only in arbitrary code or JSON; the manuscript comparison supplements that deliberately narrow parser.

Quantity	Count
Principal research packets	41
Current incoming ZIP files	26
Archive references / distinct archive payloads	71 / 44
Members of the distinct archive payloads	24,458
Recognized table occurrences	4,417
Byte-distinct tables	620
Action-preserving graph classes	446
194-state occurrences / byte forms / graph classes	97 / 15 / 8
Archive parsing failures	0

These counts include ancestors, repeated reference tables, rejected experiments and small test controls. They are not counts of correct PA consistency machines. Two disconnected synthetic tables are excluded from the graph-class count. The recorded graph signature minimizes deterministic BFS serialization over all initial controls that reach the whole working graph. It retains the read and write symbols, directions and terminal control, while forgetting the selected initial control. Thus unequal signatures imply inequivalence under a stronger relation than merely renaming states while preserving the designated start. Separate tests handle reflection and global symbol interchange.

2.2 The 194-state frontier before this work

The supplied eight graph classes contain seven constructions with implementation evidence and one documented rejection. The three original 194-state machines use the same 190-token separate-head emitter and the original Shift/MP protocol. The later cached protocol changes the source semantics, even when the arithmetic axiom family is unchanged.

Supplied design	Sites	Outcomes	Startup	Main change
PA194, 67 readers / 7 folds	3,673	4,780	58,154	Primary allocation
PA194 with A/D exchange	3,673	4,780	58,122	Physical register order
PA194, 68 readers / 8 folds	3,673	4,780	58,154	Reader association
PA194C	3,649	4,749	57,922	Cached conclusion protocol
PA194CS	3,646	4,745	57,768	Symmetric Shift comparison
PA194CZ	3,649	4,750	58,168	Known-zero scheduler jump
PA194altC	3,649	4,749	57,922	Cached, eight-fold allocation

"Outcomes" counts both branches of each decrement. Startup means an exact finite run to the entire clean source-frame tape; it is not evidence about the eventual halting behavior. Full table hashes and locators appear in `comparison/frontier194.csv`.

Five new packets principally deliver the identical PA194C table, whose SHA-256 begins `d759313c`. They should not be counted as five new machines. PA194CS is also already supplied: it omits a restore loop and exchanges the operands of a destructive equality comparison, whose normal exit clears both operands. Its extra reader is offset by a contextual fold. PA194CZ replaces source instruction 26 by a decrement of a register proved zero at that point. The alternate eight-fold cached machine appears as a *validated search input* in `pa194c_conpa.zip`; promoting it from a search directory to a headline would not make it a new construction.

The rejected eighth class illustrates the verification boundary. Its local folds and resolved source graph pass their checks, but its source addresses violate the physical return scanner's delimiter condition. Enlarging the scanner repairs the implementation at 195 states. A successful local quotient check cannot replace a complete physical execution argument.

2.3 Larger designs and contributions that survive comparison

The original PA195 uses a bundled diagonal-vacuity formula, a 191-token emitter, 3,700 source sites and 4,815 outcomes. The separate-head variant has 190 tokens and 3,673 sites. The cached 195C and bundled 195DC variants change where successful modus ponens stores its conclusion and where Shift checks for contradiction. Other 195-state results include the repaired eight-zero-scanner implementation and alternative allocations. The broader 195-state census has 26 byte forms and 16 graph classes, including known failures; those numbers do not certify every table.

The original PA196 has 3,812 source sites and 4,959 outcomes. Its important successors combine diagonal vacuity, separated emitted heads, source-tree association, physical register allocation, cached commands, and invariant based jumps. The broad 196-state census has 163 byte forms and 99 graph classes. Some experiments actually halt on blank input after 43,563 steps; others fail universal source contracts. They cannot be treated as consistency-machine improvements merely because their tables are small.

The U208 family contributes the method of recovering a merged reader's identity from neighboring tape history. The earlier positive-design catalogue identifies twenty graph classes among twenty-seven selected presentations; the much broader current census includes generated experiments and therefore

has larger counts. The U209-to-U208 history fold and the PA209-to-PA196 source redesign overlap in their inherited machinery. Their nominal state savings cannot simply be added.

The ZFC work [6] motivates finite-history reader quotients and source-level invariants. ConciseIPRA [7] motivates a different but complementary lesson: proof presentations may be changed by explicit proof translations. Neither importing an iPRA checker nor proving its consistency would by itself give the requested PA-consistency machine. Every transfer used here retains the PA refutation predicate as its logical endpoint.

2.4 Mathematical proofs and the formal endpoint

At the pinned revision the PA README explicitly says that neither machine-specific direction of the desired *iPRA* equivalence has been established for the primary PA194. Its machine-specific conditional endpoints must therefore remain conditional in this comparison. The 201-parent/194-dense quotient, zipper bridges, source coverage infrastructure, and arithmetic clock equations are valuable pieces of formal verification; they do not automatically assemble the missing internal endpoint.

This is compatible with the ordinary mathematical PA proofs in the research manuscripts. Those proofs give syntactic transformations from PA refutations to numerical-calculus refutations and back, command compilers, and positive finite physical simulations. The inherited reverse translation handles the numerical aliases at truth code zero and the empty Cantor stack, fresh diagonal binders, and coincident variable labels. It uses a uniform *proof constructor* for finite binary-set comprehension, not a truth predicate or an assumption of PA reflection. The present report states exactly which written calibration is inherited and supplies the additional finite machine and source arguments needed for its new construction. No new end-to-end Lean or Rocq theorem is claimed.

3 The source transformation

The complete source has 3,646 primitive occurrences. Each primitive is an increment, decrement with zero and positive successors, unconditional jump, or halt, acting on nine registers over $\mathbb{N}$. Two independent resolvers recover the same instruction graph from the binary source addresses: one uses address-prefix arithmetic and the other uses nested intervals. The new addresses are exactly those of PA194CS. Only the nine opcodes in the following proof change.

3.1 A local certificate for all nine zero returns

Let P be the archived PA194CS register program, and let P' be PA194ZR. A configuration consists of a source program counter and nine natural-number registers $R_0, \ldots, R_8$. The source instruction $\mathrm{dec}(r, z, p)$ leaves a zero register unchanged and jumps to z; on a positive register it subtracts one and jumps to p.

The programs have the same 3,646 instruction sites. Their only differences are the following nine replacements. All site numbers are zero-based.

site	old jump destination	new positive destination
3	5	4
9	11	10
3526	3528	3527
3532	3535	3533
3544	3546	3545
3550	3553	3551
3553	3637	3554
3627	3629	3628
3633	3637	3634

At each listed site, P' has $\mathrm{dec}(3, \text{old destination}, \text{positive destination})$.

Define the protected region

$$S = \{0, \ldots, 9\} \cup \{3523, \ldots, 3633\} \cup \{3638, \ldots, 3645\}.$$

It contains 129 sites, including all nine changed sites, and does not contain the initial source counter 25. Inspection of the complete control-flow graphs of both P and P' gives exactly three edges entering S from its complement:

$$48 \longrightarrow 3638, \qquad 49 \longrightarrow 3554, \qquad 3519 \longrightarrow 3523.$$

Each is the zero branch of a decrement of R_3. No instruction at a site in S increments R_3. The accompanying independent checker verifies these facts by enumerating every edge of both complete instruction lists; it uses no register-value bound, program simulation, or whole-program positivity-mask certificate.

Local invariant. For either program, from source counter 25 with *any* initial register store, whenever the current counter lies in S, the current value of R_3 is zero. This follows by induction on the number of source steps. The initial counter is outside S. A step entering S from outside is one of the three zero branches above and therefore establishes $R_3 = 0$. A step starting in S cannot make a zero R_3 positive, because it does not increment that register. This includes a decrement of R_3: its positive branch is then unavailable and its zero branch leaves the register unchanged.

Exact source equivalence. Start P and P' from the same arbitrary register store at source counter 25. After every number of source steps, their complete configurations are equal. Prove this by a second induction. At every unchanged site the instructions agree literally. At a changed site the local invariant gives $R_3 = 0$, so the new decrement leaves every register unchanged and takes exactly the old jump's destination. The two successors are thus identical. Consequently the source programs halt on exactly the same initial stores, after the same number of source instructions.

This is a direct elementary argument about unbounded natural-number registers. Its finite graph premises and induction can be expressed in PA. It establishes the source-level transformation only: the separate blank-tape initialization and universal physical transition contracts remain necessary to transfer the conclusion to the displayed Turing machine. In particular, the rejected version using R_6 at the eight shared returns had valid source contracts but failed blank-tape initialization, so source equivalence alone was insufficient.

3.2 Why the invariant has the right scope

The source theorem quantifies over every initial register store at source counter 25. It is stronger than a finite test of the scheduler's actual trajectory. It also avoids carrying a large whole-program positivity

analysis as a proof premise. The certificate consists of a finite region, three guarded entries, the absence of increments of one register within that region, and an exact comparison of the remaining instruction rows.

The extra positive branches in the new source are genuine parts of its syntax. They are unreachable from source counter 25 by the proved invariant. Nevertheless the physical checker verifies both outcomes of every decrement on arbitrary register frames, including those nine positive cases. There are 4,745 outcomes for PA194CS and 4,754 for PA194ZR; the increase measures additional checked contracts rather than a longer successful proof search. The source theorem gives the same number of source instructions on corresponding actual executions.

4 Binding the source to the literal machine

4.1 Canonical complete-tape frames

The logical-to-physical register permutation is

$$(R_0, \ldots, R_8) \longmapsto (6, 2, 0, 4, 3, 7, 8, 5, 1).$$

Thus R_0 occupies physical block 6, R_3 occupies physical block 4, and so on. A natural value v is represented by the block $0\,1^{v+1}$. A clean source frame contains a left sentinel, the binary source address, a zero gap, a header, nine such register blocks, and a finite nonempty list of spare blocks. The tape outside this finite word is zero. The header distance is $48 + n$ for arbitrary $n \in \mathbb{N}$; for a source address p the zero gap therefore has length $48 + n - |p|$. Every source address in this construction has length at most 48.

An increment appends a spare block, and a positive decrement may change the gap or origin. These effects are included in the full frame contract. The proof never identifies configurations merely because their first nine decoded values agree. Both sides of the tape, the head position, source address, gap and spare-list effects are tracked.

4.2 Two physical checks with different dependencies

The deterministic builder imports the frozen PA196-derived symbolic engine, checks that the register-box rows are unchanged, and verifies all 4,754 source outcomes. It performs 1,517,950 symbolic operations; its largest single outcome uses 508 operations. This executor uses inherited register-box lemmas and is identified as such in its receipt.

A separate checker imports no compiler, allocator or arithmetic-box oracle. It reconstructs four generic list-sweep lemmas from the literal transition rows, uses induction on arbitrary run lengths and arbitrary finite lists, resolves the source graph independently in two ways, and checks every source outcome. Its recorded run has 1,103,242 symbolic operations, with a maximum of 403 for one outcome. The minimum constant term in the certified physical-duration expressions is 3, which is strictly positive. The scope is all nine independent natural register values, both decrement outcomes, arbitrary natural gap parameter, and an arbitrary finite nonempty spare list. This is symbolic execution with universal loop lemmas, not bounded numerical sampling.

These checkers are executable supporting certificates. Their Python implementations are not proof-assistant kernels, and the package does not claim a fresh Lean/Rocq verification of those implementations. The mathematical interpretation of each step is finite transition execution, word equality, or induction for a stated sweep. The independent replay reduces dependence on the construction code and its arithmetic summaries.

4.3 Exact blank initialization

From native control 74, head coordinate zero and the entirely blank tape, exactly 59,816 literal transitions reach native control 84, head coordinate -2, and source instruction 25 with source address `1000000`. The logical registers are

$$(2, 0, 0, 0, 0, 0, 0, 0, 0),$$

the physical registers are $(0, 0, 0, 0, 0, 0, 2, 0, 0)$, the header distance is 48, and there are 26 one-valued spare blocks. The entire finite-support tape is compared, not a selected prefix. A second simulator implements this check independently. The new startup takes 2,048 more literal transitions than PA194CS's 57,768-step startup; no speed improvement is claimed.

The initialization obligation is indispensable. The analogous edit using R_6 at the eight shared returns passes source-level preservation and universal instruction checks but halts during the nonstandard bootstrap at step 1,361. It is rejected. The source-entry invariant begins only when a clean frame has been reached. It cannot be assumed during an arbitrary preceding tape computation.

Lemma 4.1 (Positive simulation reflects halting)**.** *For the delivered literal table, blank-tape halting is equivalent to halting of its register source from instruction 25 and the displayed initial store. The equivalence and the exact startup are formalizable in PA.*

Proof. The finite initialization reaches the exact source frame. Each source step then has a finite physical realization of positive duration to the specified full successor frame, or to the unique halt control. Iterating these realizations proves the forward source-to-tape implication.

For the converse, suppose a literal halt occurs within N transitions after the seed. If the first $N + 1$ source steps were all nonhalting, their positive realizations would give a nonhalting literal continuation of length at least $N + 1$. Determinism makes this continuation agree with the allegedly halting trace for its whole length, a contradiction. Therefore some source step halts. This argument does not assume that an infinite literal run is already known to remain within intended frames.

The finite graph identities and startup are arithmetic facts. The sweep and register trace constructors use primitive recursion on natural values, finite lists and finite source executions. Their correctness is proved by ordinary arithmetic induction, and the argument above uses bounded trace lengths and determinism. These are PA-formalizable steps. □

5 The unchanged PA calibration

This section states the logical dependency precisely and explains the composition that proves Theorem 1.1. The inherited calibration is described in the supplied manuscripts [4, 5, 3]. The relevant inherited calibration sections and cached-command audit are included under `reference/logical`; the parent symmetric-Shift proof is under `reference/parentCS/proof_symmetric.tex`.

5.1 The numerical proof calculus

Let $p(a, b) = \frac{(a+b)(a+b+1)}{2} + b$, and define

$$I(A, B) = 4p(A, B), \quad M(a, b) = 4p(a, b) + 1, \qquad U(a, B) = 4p(B, a) + 2, \quad E(a, b) = 4p(a, b) + 3.$$

The quantifier constructor deliberately reverses the Cantor arguments. Code zero is truth and code one is the contradiction $M(0, 0)$. Write $\neg A = I(A, 1)$, with conjunction and existential quantification as the

usual classical abbreviations. The six ordinary axiom families are

$$\begin{aligned} h_1(D,H) &= (\exists x_D\, U(D,H)) \to H,\\ h_2(D,A,H) &= U(D,U(A,H)) \to U(A,U(D,H)),\\ h_3(D,A,H) &= M(H,D) \to (E(H,A) \to M(A,D)),\\ h_4(D,A,H) &= E(H,D) \to (M(A,H) \to M(A,D)),\\ h_5(D,H) &= \exists x_D E(H,D),\\ h_6(D,A,H) &= U(D,A) \to (U(D,A\to H) \to U(D,H)). \end{aligned}$$

Retain the restricted propositional family

$$\rho_8(D,A,H,u) = ((H\to u)\to D)\to((D\to H)\to(A\to H)), \qquad u \le D,$$

and the rules of modus ponens and generalization on derivations without undischarged hypotheses. Define

$$\begin{aligned} \Gamma(B) &= B \to U(B,B),\\ V(a,b,c) &= M(a,b) \to U(c,M(a,b)),\\ P(a,b,c) &= c \to M(c,b),\\ S(a,b,c) &= U\big(b,(M(a,b)\to P(a,b,c))\wedge(M(c,b)\to c)\big),\\ Q(a,b,c) &= U(c,S(a,b,c)\to\neg P(a,b,c)),\\ G(a,b,c) &= \neg(V(a,b,c)\to Q(a,b,c)). \end{aligned}$$

The separate-head calculus adds every $\Gamma(B)$ and the guarded $G(a,b,c)$ for $c \neq a,b$. A failed guard emits zero. These are numerical formula codes, so notation such as the occurrence of c as a formula inside P is intentional.

The inherited calibration supplies PA-verifiable, primitive-recursive transformations proving

$$\mathrm{PA} \vdash \big(\exists e\ \mathrm{Prf}_{\mathrm{PA}}(e,\ulcorner 0=1\urcorner) \leftrightarrow \exists d\ \mathrm{Prf}_s(d,1)\big). \tag{C}$$

Its forward direction reuses the PA196 refutation compiler with explicit replacements for equality vacuity and neighborhood suppliers. Diagonal freshness is justified by the inequality that every variable label in a formula of code B is strictly below B. The reverse direction translates $M(a,b)$ to the arithmetic bit relation, E to equality and U to universal quantification. PA proves that a number cannot contain itself as a binary-position element. It constructs proofs of bounded binary-set comprehension by induction on the bound, syntactically inserting the requested defining formula. This is a proof-producing construction and does not invoke PA soundness or reflection. The code-zero alias $I(0,0) = 0$ is handled by a constructor *equivalence*, rather than an incorrect syntactic identity.

The relevant inherited sections are `reference/logical/original_pa195_calibration.tex` and `reference/logical/separate_head_calculus.tex`. The command compilation below is supplied by `reference/logical/cached_command_audit.tex`.

5.2 Cached commands and the refutation compiler

Let $[a;w] = p(a,w)$, with total Cantor projections hd and tl, and write the command store as (t,w,d,h). The unchanged commands of PA194CS and PA194ZR are

$$\mathsf{S}(t,w,d,h) = \begin{cases} \mathsf{halt}, & h = 1,\\ (h,\mathrm{tl}(w),d,\mathrm{hd}(w)), & h \neq 1, \end{cases}$$

and, with $v = d + x$,

$$\mathsf{MP}_x(t, w, d, h) = \begin{cases} (0, w, 0, v), & I(t, v) = h, \\ (0, w, v, 0), & I(t, v) \neq h. \end{cases}$$

For $D = d + x$, $H = h + y$, $A = c$, AX pushes, in order,

$$t,\ U(d, h),\ \rho_8(D, A, H, d),\ h_1, h_2, h_3, h_4, h_5, h_6, \Gamma(A),$$

leaves the guarded $G(H, D, A)$ on top, and clears both caches. The stack has ten newly consed entries including the saved top. Every newly generated formula except $U(d, h)$ has residue zero modulo four; $U(d, h)$ has residue two. Thus none of the generated formulas is code one; the saved old top t is unrestricted.

For completeness the principal command words are

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}_{0,0,0}; \mathsf{S}^9; \mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}_{0,0,0}; \mathsf{S}^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; \mathsf{S}^{10}; \mathsf{MP}_0. \end{aligned}$$

Push saves the top and clears the visible proof registers. Pop preserves the top and removes one stack entry, except that it deliberately halts if the preserved top is one. Discard clears the top when the caches are zero. The numerical alias $[0; 0] = 0$ is valid throughout; these are identities of total Cantor projections, not assumptions about an injective finite list representation.

If q_j is the jth newly pushed entry counted from the top, ordinary head selection for $1 \leq j \leq 7$ uses

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathsf{Pop}^{9-j}.$$

Here $q_1 = \Gamma(A)$ and $(q_2, \ldots, q_7) = (h_6, \ldots, h_1)$. The separate selectors for ρ_8 and the gate, and the generalization word, are written out in the included cached-command audit.

For modus ponens, the suffix $\mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{S}$ acts on $(0, [a, I(a, b), 0; w], 0, 0)$. It either detects code one early or returns $(b, w, 0, 0)$. The explicit dummy zero prevents consuming the first underlying stack entry. Recursion on the proof tree then compiles every calculus proof into a finite word which either halts or returns its conclusion with the original stack and zero caches. In particular a proof of one yields a command word that halts from *every* incoming store after a clearing AX prefix. That stronger property makes the unchanged exhaustive scheduler sufficient even though earlier candidate trials may leave nonempty stores.

Conversely, annotate the visible top, cached head and every requested canonical stack entry by calculus proofs. AX supplies axioms, a guarded axiom or truth; MP applies the rule or returns truth; Shift transports proofs. A halting Shift exposes a proof of one. A fixed truth proof handles the zero tail. All updates are finite proof operations. Together with (C) and the scheduler this gives the inherited PA calibration of the cached source. The symmetric Shift optimization has the same complete command effects because its destructive equality test clears both operands on normal exit.

5.3 Composition of the new theorem

Let P_{CS} be the archived source and P_{ZR} the new source. Their initial source configuration is counter 25 with register vector $(2, 0, \ldots, 0)$. The protected-region argument proves in PA that their complete source traces agree. Lemma 4.1 binds the literal PA194ZR table to P_{ZR}. The inherited syntactic calibration binds P_{CS} to a PA refutation. Composing these equivalences gives

$$\mathrm{PA} \vdash \mathrm{Halt}(M_{194\mathrm{ZR}}, \mathrm{blank}) \leftrightarrow \exists e\ \mathrm{Prf}_{\mathrm{PA}}(e, \ulcorner 0 = 1 \urcorner).$$

Negating both sides in classical PA proves Theorem 1.1. All newly required arguments are finite graph checks, arithmetic induction and complete finite-tape simulations. This report does not upgrade an unfinished iPRA formal endpoint into a completed theorem.

6 Attempts to cross the 194-state frontier

No delivered search produced a verified machine with fewer than 194 working states. This is an experimental outcome, not a lower bound. The searches separate syntactic size estimates, local tape-context conditions, complete arbitrary-register contracts, and the actual blank initialization. Passing an earlier gate is never reported as passing a later one.

6.1 Compiler and axiom-presentation searches

The compiler searches vary the binary association of the axiom-emitter word, its shared reader dictionary, and context-sensitive reader folds. They also test permutations of emitted heads, parameter roles, selected macro expansions, and the new zero-return control. The frozen trusted record accounts for 90,819,000 mutation iteration slots. A slot is not necessarily a distinct grammar or a complete candidate machine, and most slots never reach whole-machine validation. The best resulting total state-count score remains 194.

Two logical variants exchange the antecedents of h_3 or h_6. The exchange transformation has an explicit propositional proof and is a legitimate candidate for preserving the proof calculus. Its compiler scores nevertheless remain above 194 in these runs. The accompanying exchange-template checker is retained as exploratory work; neither variant is substituted into the delivered machine.

A dynamic-normalization defect was discovered during exploration: normalization preserved the emitted word immediately but reused leaf position identifiers that a later mutation treated as distinct. An apparent sub-194 result from the affected experiment was discarded. The corrected optimizer renumbers the leaves and checks exact word identity after normalization. The trusted accounting excludes all runs affected by that defect, unlogged interrupted tails, and runs based on the blank-invalid R_6 return variant. The final two bounded runs were stopped at logged prefixes, with 4,743,000 and 4,503,000 accounted slots. This provenance is necessary to interpret the negative record honestly.

6.2 Physical reductions of the accepted machine

The final PA194ZR table was also subjected to direct reduction attempts. Native control 302 remains reachable through the zero branch of control 113 even after the nine source returns have been redirected. The successful source transformation therefore does not by itself delete 302.

Bounded search	Trials	Result
Mergers involving 302, selecting old transition actions	676	No complete-contract success
Alternative actions for $(113, 0)$ after deleting 302	776	No complete-contract success
Generalized pair folds on radius-six contexts	28,755,456	No context-valid reduction
302-specific folds on radius-eight contexts	297,984	No context-valid reduction

The context collections contain 272,471 and 3,129,859 contexts respectively. These are finite inductive abstractions of tape behavior, and a negative result can exclude only the tested actions under the

specified abstraction. It is not a proof that a different simulation relation, source tree, scanner, or arithmetic kernel cannot save a state.

Earlier scanner experiments shorten the seven-zero delimiter test to six, redirect scanner states, or merge scanner controls. They fail universal contracts. The six-zero version fails 95 contracts: delimiter safety cannot be inferred merely from a tally of source-address substrings. The record includes the important R_6 negative control: all source contracts pass, but the literal blank run halts at step 1,361. It is a rejected design, never an alternative consistency machine.

6.3 What remains open

The accepted construction changes the shared return convention while keeping the same total state count. Its two row changes could provide a different starting point for future reader allocation or scanner work, but no further saving is established here. A successful smaller machine would still need a complete source binding, arbitrary-frame physical contracts, exact blank initialization, and a PA refutation calibration. The present evidence neither proves minimality at 194 nor rules out a substantially different design.

7 Reproduction and artifact guide

Python 3.10 or later and its standard library suffice for the frozen construction and all delivered primary checks. No network, optimizer, SAT solver, Lean installation or Rocq installation is required to reconstruct the table. Run without Python's `-O` option because assertions are certificate checks.

```
python3 -B build_verify.py
python3 -B verification/check_source_equivalence.py \
  --original reference/parentCS/machine/pa194CS.instructions.json \
  --modified source/instructions.json \
  --output proof/independent_source_invariant.json
python3 -B verification/check_literal.py
python3 -B comparison/check_novelty.py \
  --candidate machine/pa194zr.tm --start 0 \
  --census comparison/census_current/census.json \
  --table-dir comparison/census_current/tables \
  --output comparison/novelty_replayed.json
```

The first command reconstructs every table, source list, local proof certificate and exact startup, compares them with the frozen files, and checks the inherited-engine universal contracts. The separate literal checker rederives its list lemmas and checks all source outcomes, so it can take several minutes. Recorded receipts are included. Discovery searches are unnecessary for reproducing the accepted result.

The complete report builds with two ordinary `pdflatex` runs:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error pa194zr.tex
pdflatex -interaction=nonstopmode -halt-on-error pa194zr.tex
```

A source-provenance manifest records the pinned commit and imported files. `SHA256SUMS` covers the delivered contents except itself. The literal transition table and its digest above define the machine; search filenames and incidental native state numbers do not define a second initial-input convention.

References

[1] ProveIt contributors, *PA consistency machines and delivered research packets*, revision `17f92105e2bc4317feae6c219cb2143437b482ff`. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/PA.

[2] ProveIt contributors, *Incoming research archives*, same pinned revision. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/docs/incoming.

[3] *Two new 194-state PA consistency machines*, supplied `pa194_new_designs_2026-11-13.zip`; PA194CS construction, symmetric-Shift proof and independent verification. Parent construction retained in `reference/parentCS`.

[4] *PA195* and the separate-head follow-up: ordinary syntactic PA calibration and cached-command compiler, supplied manuscripts retained under `reference/logical`.

[5] James Chang, *BusyBeaver: PA196 development*, imported snapshot under `PA/jamesyc-upstream/pa196`; in particular `PA196/Logic/Basic.lean`, `ImplicationBasis.lean`, `Deduction.lean` and the relational refutation suppliers. https://github.com/jamesyc/BusyBeaver.

[6] ProveIt contributors, *ZFC consistency-machine research*, finite-history quotients and source invariants. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/ZFC.

[7] ProveIt contributors, *ConciseIPRA*, proof presentations and explicit translations. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA.

A The complete 194-state transition table

State zero is initial; label 194 is the distinct halt control. An entry wDq writes w, moves D, and enters q. Each printed state has both read-symbol transitions. This table is generated directly from the hashed five-column file, without manual transcription.

State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R2	97	0L112	1R113
1	1R3	1R1	98	0R114	0L14
2	0R0	1R2	99	0R115	1R91
3	0R4	0R1	100	0R115	1R106
4	1R5	1L6	101	0R116	0L14
5	1L7	1L8	102	0R68	1L66
6	0L7	1L6	103	0R117	1R118
7	0L9	1L6	104	0R89	1R104
8	0L10	0L11	105	0R14	1R105
9	1R12	0L9	106	0L112	1L33
10	0L9	0L10	107	0R67	1R119
11	1L8	1L11	108	0R63	1R108
12	0R12	0R13	109	0R120	1L121
13	0L14	0R13	110	0R122	1R123
14	1L15	0L15	111	0R46	0L14
15	0L15	1L16	112	0R86	1R96
16	0L17	1L16	113	0R112	1L91
17	0L18	1L16	114	0R124	1R124
18	0L19	1L16	115	0L60	0R83
19	0L20	1L16	116	0R62	0R89
20	0L21	1L16	117	0L81	1R125
21	0L22	1L16	118	0R126	1R127
22	0L23	1L16	119	0R47	1R85
23	0R23	1R24	120	0R55	0R108
24	0R25	1R26	121	0R35	1R33
25	0R27	1R28	122	0R95	1R111
26	0R29	1R30	123	0R75	1R74
27	0R31	1R194	124	0R46	1R46
28	0L32	1R33	125	0R128	1R129
29	0R34	1R35	126	0R32	1R64
30	0L33	1L36	127	0R31	1R109
31	0R37	1R38	128	0R130	1R131
32	0R39	1R40	129	0R132	1R133
33	0R41	1R42	130	0R134	1R135
34	0R43	1R44	131	0R136	1R137
35	0R45	1R36	132	0R138	1R139
36	0R46	1R47	133	0R140	1R141
37	0R48	1R41	134	0R93	1R121
38	0R49	1R50	135	0R142	1R124
39	0R51	1R52	136	0R143	1R144
40	0R53	1R54	137	0R145	1R146
41	0R55	0R56	138	0R134	1R147
42	0R57	0L14	139	0R46	1R148
43	0R58	1R59	140	0R149	1L131
44	0R60	1R49	141	0R150	1R151
45	0R55	0R61	142	0R72	1R40
46	0R62	0R63	143	0R28	1R152

State	Read 0	Read 1	State	Read 0	Read 1
47	0L64	1L65	144	0L153	1L154
48	0R45	1R50	145	0L129	1R155
49	0L66	0L15	146	0R156	1R157
50	0R67	1R68	147	0R149	1R158
51	0R69	1R70	148	0R159	1R46
52	0R71	1R72	149	0R93	1R160
53	0R73	1R74	150	0R161	1R162
54	0R75	1R76	151	0R163	1R68
55	0L77	0R1	152	0R164	1R165
56	0R61	1R56	153	0R166	1R157
57	0L69	1L35	154	0R167	1R156
58	0R78	1R79	155	0R135	1R130
59	0R51	1R80	156	0R147	1R40
60	0L81	1R82	157	0R134	1R168
61	0R83	1R61	158	0R40	1L169
62	0L77	0R2	159	0L170	1R171
63	0R84	1R63	160	0R102	1R80
64	0R30	1R85	161	0R172	1R170
65	0R86	1R87	162	0R173	1R174
66	0R35	1R88	163	0R175	1R176
67	0R41	0L13	164	0R177	1R178
68	0R55	0R89	165	0L179	1L180
69	0R90	1R65	166	0R149	1R140
70	0R47	1R91	167	0R165	1R181
71	0R41	1R92	168	0L130	1R46
72	0R93	1R94	169	0R182	1R135
73	0R95	1L96	170	0R183	1R145
74	0R47	1R73	171	0R158	1R130
75	0R88	1R97	172	0R139	1R165
76	0R47	1R98	173	0R158	1R184
77	1R77	1L15	174	0R185	1R102
78	0R90	0L7	175	0R31	1R49
79	0R99	1R100	176	0R186	1R175
80	0R41	1R101	177	0R134	1R142
81	0R102	1R85	178	0R32	1R187
82	0R103	0L67	179	0R144	1R188
83	0R104	1R83	180	0R189	1R190
84	0R105	1R84	181	0R40	1R191
85	0R106	0L14	182	0R168	1R40
86	0R62	0R104	183	0R192	1R46
87	0R107	0L85	184	0R153	1R146
88	0R30	1R57	185	0R165	1R180
89	0R108	1R89	186	0R119	1R109
90	0L60	0R84	187	0R193	1R159
91	0R90	1L109	188	0R169	1R154
92	0R97	0L14	189	0R158	1R192
93	0R64	1R110	190	0R179	1R182
94	0R109	1R71	191	0R182	1R40
95	0L60	0R105	192	0R149	1R168
96	0L59	1R111	193	0R72	1R144

B Archive-by-archive ledger

The following ledger accounts for all forty-one principal packets. Historical descriptions summarize the supplied manuscripts; they do not assert that every historical experiment was rerun. Detailed hashes, locators and status qualifications are in the CSV/JSON comparison files.

Packet	Contribution and status
pa_unary_power_cover.zip	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
U208_ConPA_Conditional_Context_2026-10-29.zip	U208 by a conditional tape-context reduction. States a 208-state candidate machine/u208.tm.
U208_ConPA_Entry_Context_2026-10-29.zip	U208 by eliminating a state using entry context. States a 208-state machine machine/u208.tm.
U208_ConPA_History_2026-10-29.zip	U208 by a finite-history decoder fold. States a 208-state candidate machine/u208.tm.
U208_ConPA_2026-10-29.zip	U208 by a three-cell fold. States a 208-state candidate machine/u208.tm.
u208_research_2026-10-29.zip	U208 research release. States two 208-state machines, machine/u208.tm and machine/variant_plain/u208.tm.
Below_U208_Research_2026-10-29.zip	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
u208_comparison_research_2026-10-29.zip	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
ConPA_below208_research_2026-10-29.zip	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.
pa_consistency_research_2026-10-29.zip	Audit and new source reductions. States that no validated machine below 208 states was obtained.
pa208_research_2026-10-29.zip	Comparison and research below 208 states. States that it does not establish a valid machine below 208.
pa_tm_research_2026-11-13.zip	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
pa_consistency_research_2026-11-13.zip	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
conpa_tm_comparison_and_reduction_search.zip	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
pa196_research_2026-11-13.zip	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
pa_consistency_research.zip	Research and verification packet. States that no machine below 196 states has been validated.
pa_tm_comparison_and_reductions.zip	Comparison and new rewrites, a research snapshot.
pa_tm_extended_research_2026-11-13.zip	Comparison and refined reduction search. States that the smallest accepted count is 196.
conpa_tm_comparison_and_search.zip	Comparison and a search below 196. States that none was obtained; files named 195 are rejected or diagnostic candidates.
pa_consistency_reduction.zip	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
pa_consistency_comparison_2026_11_13.zip	Comparison and new reduction experiments. States that nothing below 196 was obtained.
PA_consistency_TM_research_2026-11-13.zip	Comparison and new source reductions. States that no machine below 196 is claimed.
pa_consistency_research_2026-11-13(2).zip	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
conpa_tm_study_2026-11-13.zip	Comparison and reduction study. States that nothing below 196 was obtained.
pa_below196_research.zip	Comparison and reduction experiments. States that nothing below 196 was obtained.
pa_consistency_comparison_and_new196_2026-11-13.zip	Comparison and reduction research. States that it contains no accepted machine below 196.

Packet	Contribution and status
pa_tm_research_2026-11-13(2).zip	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
pa_consistency_cross_archive.zip	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
pa195_conpa_2026-11-13.zip	PA195. States a 195-state machine pa195/pa195.tm and an article with a PA proof of the equivalence.
pa_consistency_reduction_audit_2026-11-14.zip	Comparison and reduction audit. States nothing below 196 and no new theorem.
pa_consistency_research_2026-11-14.zip	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.
PA195_ConPA_research_2026-11-14.zip	A 195-state machine. States the primary machine_195S.tm and a second table machine_195C.tm.
pa195_followup_research.zip	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.
pa194_conpa_2026-11-13.zip	A 194-state machine. States the table construction/tables/pa194.tm and a proof of the equivalence in PA.
pa194_conpa_2026-11-13(2).zip	PA194. States a 194-state machine pa194/pa194.tm and an article with a PA proof of the equivalence.
pa195_diagonal_combined_2026-11-13.zip	PA195DC. States a different 195-state machine.
pa194C_conpa_2026-11-13(2).zip	PA194C cached controller composed with primary 194-state decoder. Principal graph is the same d759313c machine delivered by the other C packets; mathematical PA calibration plus universal literal-tape certificate, no new kernel endpoint.
pa194c_conpa.zip	PA194C; also validated eight-fold cached-controller search input, native a509952c. Same principal C graph. Eight-fold C is distinct and passed starting-point source/runtime checks; it cannot be counted as a new fallback.
pa194c_conpa_2026-11-13(1).zip	PA194C; also invariant-based zero-jump PA194CZ, dense b713e734. Same principal C graph. CZ builder plus 263,244-pair positivity invariant; no separate fresh literal-tape independent replay for CZ.
pa194c_conpa_2026-11-13.zip	PA194C; restricted physical/scanner and 27,851,976-proposal compiler searches. Same principal C graph; searches did not produce sub-194 and are not lower bounds.
pa194_new_designs_2026-11-13.zip	PA194C plus PA194CS symmetric destructive comparison, dense 1c71002e. CS removes 3 primitive sites and trades a new reader for a fold of state 41 into 330; 3,646 sites, 4,745 outcomes, 57,768 startup steps; independent literal-tape checks supplied.